

The local approach to causal inference under network interference

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We propose a new nonparametric modeling framework for causal inference when outcomes depend on how agents are linked in a social or economic network. Such network interference describes a large literature on treatment spillovers, social interactions, social learning, information diffusion, disease and financial contagion, social capital formation, and more. Our approach works by first characterizing how an agent is linked in the network using the configuration of other agents and connections nearby as measured by path distance. The impact of a policy or treatment assignment is then learned by pooling outcome data across similarly configured agents. We demonstrate the approach by deriving finite-sample bounds on the mean-squared error of a k -nearest neighbor estimator for the average treatment response as well as proposing an asymptotically valid test for the hypothesis of policy irrelevance. We illustrate the empirical applicability of our method with simulations and an application to social capital formation.

KEYWORDS. Causal inference, network interference, social networks, nonparametric estimation.

JEL CLASSIFICATION. C14, C21.

Economists are often tasked with predicting outcomes under a counterfactual policy or treatment assignment. In many cases, the counterfactual depends on how the agents are linked in a social or economic network. A diverse literature on treatment spillovers, social interactions, social learning, information diffusion, disease and financial contagion, social capital formation, and more, approaches this problem from a variety of specialized, often highly-parametric frameworks (see broadly [Graham \(2011\)](#),

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Blume, Brock, Durlauf, and Ioannides (2010), de Paula (2016), Athey and Imbens (2017), Jackson, Rogers, and Zenou (2017), Bramoullé, Djebbari, and Fortin (2020)). In this paper, we propose a new framework for causal inference that accommodates many such examples of network interference.

Our main innovation is a nonparametric modeling approach for sparse network data based on local configurations. Informally, a local configuration refers to the features of the network (the agents, their characteristics, treatment statuses, and how they are connected) nearby a focal agent as measured by path distance. The idea is that these local configurations index the different ways in which a policy or treatment assignment can impact the focal agent's outcome under network interference.

This approach generalizes a developed literature on spillovers and social interactions in which the researcher specifies reference groups or an exposure map that details exactly how agents influence each other (see, for instance, Manski (1993, 2013), Hudgens and Halloran (2008), Aronow and Samii (2017), Vazquez-Bare (2017), Leung (2019), Arduini, Patacchini, and Rainone (2020), Sävje (2021), Qu, Xiong, Liu, and Imbens (2021)). One limitation of this literature is that the effect of a policy or treatment assignment is generally sensitive to how the researcher models the dependence. For example, in the spillovers literature, it is often assumed that agents respond to the average treatment of their peers, while in the diffusion literature agents may be informed or infected by any peer. When the researcher is uncertain as to exactly how agents influence each other, misspecification can lead to inaccurate estimates and invalid inferences about the impact of the policy or treatment assignment of interest.¹

Another limitation of this literature is that it does not generally consider policies that change the structure of the network. Network-altering policies are common in economics. Examples include those that add or remove agents, or connections between agents, from the community (see, for instance, Ballester, Calvó-Armengol, and Zenou (2006), Azoulay, Graff Zivin, and Wang (2010), Donaldson and Hornbeck (2016), Lee, Liu, Patacchini, and Zenou (2021)). Such policies may be difficult to evaluate using standard frameworks, which typically focus on the reassignment of treatment to agents keeping the network structure fixed.²

Our methodology addresses these limitations by using local configurations to model network interference. Intuitively, we use the space of local configurations as a “network sieve” that indexes the distribution of agent-specific outcomes associated with a given policy or treatment assignment. A contribution of our work is to formalize this local approach and apply it to causal inference under network interference.

The use of local configurations in economics was proposed by de Paula, Richards-Shubik, and Tamer (2018), Anderson and Richards-Shubik (2019) and related to that of

¹The frameworks of Leung (2019), Sävje (2021) are misspecification-robust in the sense that their inference procedures may still be valid even if the model of interference used to define the estimand is misspecified. Their procedures do not address the issue we highlight here that an estimand based on a misspecified model may fail to accurately describe the impact of the policy of interest. We discuss this issue in more detail in our note Auerbach, Auerbach, and Tabord-Meehan (2024).

²An example of a policy where an agent is removed from a social network is incarceration, where an agent is forcibly removed from society and put in jail.

ego-centered networks in sociology (see generally [Wasserman and Faust \(1994\)](#)). In their work, local configurations index moment conditions that partially identify the parameters of a strategic network formation model. The researcher has the flexibility to choose the configurations used for this task and can restrict attention to a small number that occur frequently in the data. In our setting, local configurations correspond to fixed counterfactual policies. It is usually the case that no exact instances of a given policy appear in the data and so we substitute outcomes associated with similar but not exactly the same configurations. Formalizing this procedure requires new machinery, which we introduce building on work by [Benjamini and Schramm \(2001\)](#).

We demonstrate our local approach with applications to two causal inference problems. In both problems, the researcher starts with a status-quo policy as described by one local configuration and is tasked with evaluating the impact of an alternative policy as described by another local configuration. The researcher has access to data from multiple networks corresponding to a partial or stratified interference setting. Such designs are common in education, industrial organization, labor, and development economics where the researcher may collect network data on multiple independent schools, markets, firms, or villages.

The first problem is to estimate average or distributional policy effects/treatment response. For instance, the status-quo policy may be given by a particular network structure and the new policy may be one in which a key agent is removed. The policy effect to be estimated is the expected change in outcome for one or more agents. We propose a k -nearest neighbors estimator for the policy effect and provide nonasymptotic bounds on mean-squared error building on work by [Döring, Györfi, and Walk \(2017\)](#). We also provide sufficient conditions for the estimator to be asymptotically normal, which we use to construct confidence intervals for the policy effect in the usual way.

The second problem is to test policy irrelevance/no treatment effects. For instance, the status-quo policy may be given by a particular network structure where no agents are treated and the new policy may keep the same connections between agents but have every agent treated. The hypothesis to be tested is that both policies are associated with the same distribution of outcomes for one or more agents. We propose an asymptotically valid randomization test for this hypothesis building on work by [Canay and Kamat \(2018\)](#).

We illustrate our local approach with simulations and an empirical application to social capital formation. In our simulations, which can be found in Supplemental Appendix Section C ([Auerbach, Guo, and Tabord-Meehan \(2026\)](#)), we find using two outcome models, three network formation models, and eight network configurations that our test for policy irrelevance/no treatment effects has, for many specifications, a rejection rate of at least 50–80% for empirically reasonable medium-sized samples when the hypothesis to be tested is false.

Section 1 defines a local configuration. Section 2 incorporates our definition of a local configuration into an econometric model with network interference. Section 3 describes applications to estimating policy effects and testing policy irrelevance. Section 4 contains an empirical illustration evaluating the impact of social network structure on favor exchange in the setting of [Jackson, Rodriguez-Barraquer, and Tan \(2012\)](#). Section 5

concludes. Proof of claims, simulation evidence, and other details can be found in a Supplemental Appendix (Auerbach, Guo, and Tabord-Meehan (2026)).

1. LOCAL CONFIGURATIONS

In this section, we first provide an informal description of a local configuration along the lines of de Paula, Richards-Shubik, and Tamer (2018). We then give a formal definition.

1.1 Informal description

Intuitively, agent i 's local configuration refers to the agents within path distance r of i , their characteristics, and how they are connected. Larger values of r are associated with more complicated configurations, which give a more precise picture about how i is connected in the network. This idea is illustrated in Figure 1.

Panel (a) depicts twelve agents connected in an unweighted and undirected network with a binary individualized treatment. Agents are either assigned to the treatment (red square nodes) or the control (blue circle nodes). Panel (b) depicts the local configurations of radius 1 for agents 1 and 2. They are both equivalent to a wheel with the focal untreated agent in the center and three other agents on the periphery, one of which is treated. Panel (c) depicts the local configurations of radius 2 for agents 1 and 2. They are both equivalent to a ring between five untreated agents (one of which is the focal agent)

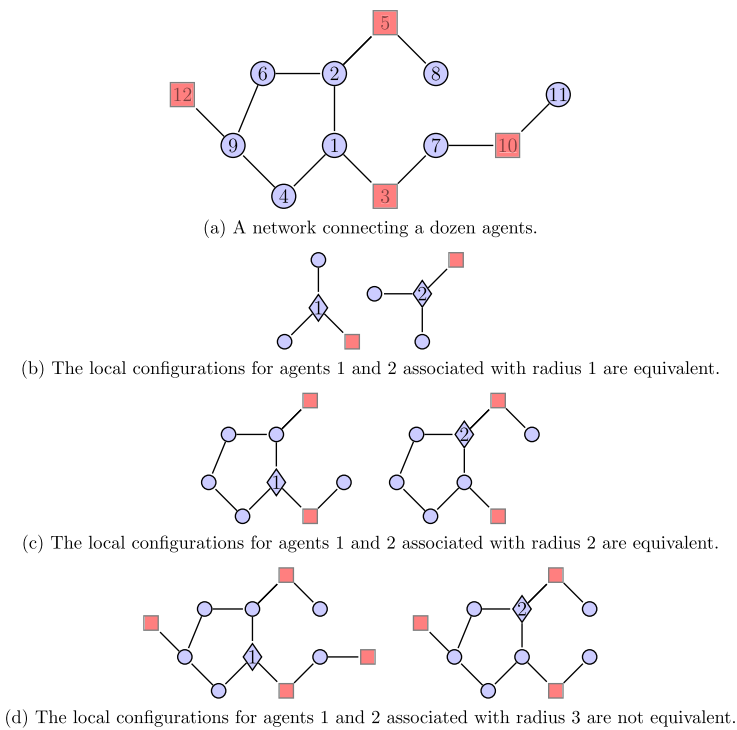


FIGURE 1. Illustration of local configurations.

where the focal agent is also connected to a treated agent who is connected to an untreated agent and another agent in the ring adjacent to the focal agent is connected to a treated agent. Panel (d) depicts the local configurations of radius 3 for agents 1 and 2. They are not equivalent because the local configuration for agent 1 contains four treated agents while the local configuration for agent 2 contains only three treated agents.

In this way, one can describe the local configurations for any choice of agent and radius. Since the diameter of the network (the maximum path distance between any two agents) is 7, any local configuration of radius greater than 7 will be equal to the local configuration of radius 7. However, for networks defined on large connected populations, increasing the radius of the local configuration typically reveals a more complicated network structure.

Following [Benjamini and Schramm \(2001\)](#), we call the infinite-radius local configuration a rooted network. We now provide a formal definition.

1.2 Formal definition

We formally define the local configuration from Section 1.1. To do this, we first review some terminology and notation from the network theory literature. We then define the space of rooted networks building on [Benjamini and Schramm \(2001\)](#), [Aldous and Steele \(2004\)](#).

1.2.1 Terminology and notation A countable population of agents indexed by $\mathcal{I} \subseteq \mathbb{N}$ is linked in a weighted and directed network. The weight of a link from agent i to j is given by $D_{ij} \in \mathbb{Z}_+ \cup \{\infty\}$. Each agent is also assigned a treatment $T_i \in \mathbb{R}$. The population vector of treatment assignments is denoted $\mathbf{T} = \{T_i\}_{i \in \mathcal{I}}$ and $\mathcal{T}$ is the set of all possible assignments.

We explicitly allow for weighted networks because they are used in many of our motivating examples. They might measure, for example, the amount of physical distance between students, number of social connections surveyed between villagers, or degree of collaboration between researchers. The matrix D indexed by $\mathcal{I} \times \mathcal{I}$ with D_{ij} in the ij th entry is called the adjacency matrix. We take the convention that larger values of D_{ij} correspond to weaker relationships between i and j . We suppose that $D_{ij} = 0$ if and only if $i = j$. When the network is unweighted (agent pairs are either linked or not) $D_{ij} = 1$ denotes a link and $D_{ij} = \infty$ denotes no link from i to j .

A path from i to j is a finite ordered set $\{t_1, \dots, t_L\}$ whose entries take values in $\mathbb{N}$, $t_1 = i$, $t_L = j$, and $L \in \mathbb{N}$. The length of the path $\{t_1, \dots, t_L\}$ is given by $\sum_{s=1}^{L-1} D_{t_s t_{s+1}}$. The path distance from i to j denoted $\rho(i, j)$ is the length of the shortest path from i to j . That is,

$$\rho(i, j) := \inf_{\substack{\{t_1, \dots, t_L\} \\ \text{s.t. } t_1 = i, t_L = j}} \sum_{s=1}^{L-1} D_{t_s t_{s+1}}.$$

If the path distance from i to j is finite, we say that i is path connected to j . For any $i \in \mathcal{I}$ and $r \in \mathbb{Z}_+$, agent i 's r -neighborhood $\mathcal{I}_i(r) := \{j \in \mathcal{I} : \rho(i, j) \leq r\}$ is the collection of agents

within path distance r of i . $N_i(r) := |\mathcal{I}_i(r)|$ is the size of agent i 's r -neighborhood (i.e., the number of agents in $\mathcal{I}_i(r)$). For any agent-specific variable (such as an outcome or treatment assignment) $\mathbf{W} := \{W_i\}_{i \in \mathcal{I}}$, $W_i(r) := \sum_{j \in \mathcal{I}} W_j \mathbb{1}\{\rho(i, j) \leq r\}$ is the r -neighborhood count of $\mathbf{W}$ for agent i . It describes the partial sum of $\mathbf{W}$ for the agents in $\mathcal{I}_i(r)$. $\mathcal{I}_i(0) = \{i\}$, $N_i(0) = 1$, and $W_i(0) = W_i$ since $D_{ij} = 0$ if and only if $i = j$.

We assume that the network (adjacency matrix) is *locally finite*. That is, for every $i \in \mathcal{I}$ and $r \in \mathbb{Z}_+$, $N_i(r) < \infty$. In words, the assumption is that every r -neighborhood of every agent contains only a finite number of agents. The assumption is implicit in much of the literature on network interference (including the examples in Section 3.2 below) where the researcher observes all of the relevant dependencies between agents in finite data. We do not assume a uniform bound on the size of the r -neighborhoods. The set of all locally finite adjacency matrices is denoted $\mathcal{D}$.

1.2.2 Rooted networks For an adjacency matrix $D \in \mathcal{D}$ and treatment assignment vector $\mathbf{T} \in \mathcal{T}$, a network is the triple $(\mathcal{I}, D, \mathbf{T})$ where $\mathcal{I}$ is the vertex set and D is the weighted edge set. A rooted network $G_i = G_i((\mathcal{I}, D, \mathbf{T}), i)$ is the triple $(\mathcal{I}, D, \mathbf{T})$ with a focal agent $i \in \mathcal{I}$ called the root. Informally, G_i is the network $(\mathcal{I}, D, \mathbf{T})$ “from the point of view” of i . In this paper, we will often use the abbreviated notation $G_i = G_i(D, \mathbf{T})$ when the population $\mathcal{I}$ is clear. We will also use notation like $\mathcal{I}(g)$, $D(g)$ and $\mathbf{T}(g)$ to refer to the vertex set, weighted edge set, and set of treatment assignments associated with a given rooted network g .

For any $r \in \mathbb{Z}_+$, G_i^r is the subnetwork of $G_i((\mathcal{I}, D, \mathbf{T}), i)$ induced by the agents within path distance r of i as measured by path distance ρ . Formally, $G_i^r := ((\mathcal{I}_i(r), D_i(r), T_i(r)), i)$, where $\mathcal{I}_i(r) := \mathcal{N}_i(r) := \{j \in \mathcal{I} : \rho(i, j) \leq r\}$, $D_i(r) := \{D_{jk} : j, k \in \mathcal{I}_i(r)\}$, and $T_i(r) := \{T_j \in \mathbf{T} : j \in \mathcal{I}_i(r)\}$. We say G_i^r is the rooted network G_i truncated at radius r . The rooted network formalizes the idea of a local configuration, which we described previously in Section 1.1.

For any $\varepsilon \geq 0$, two rooted networks G_{i_1} and G_{i_2} are ε -isomorphic (denoted $G_{i_1} \simeq_\varepsilon G_{i_2}$) if all of their r -neighborhoods are equivalent up to a relabeling of the nonrooted agents, but where treatment assignments are allowed to disagree up to a tolerance of ε . Formally, $G_{i_1} \simeq_\varepsilon G_{i_2}$ if for any $r \in \mathbb{Z}_+$ there exists a bijection $f : \mathcal{I}_{i_1}(r) \leftrightarrow \mathcal{I}_{i_2}(r)$ such that $f(i_1) = i_2$, $D_{jk} = D_{f(j)f(k)}$, for any $j, k \in \mathcal{I}_{i_1}(r)$, and $|T_j - T_{f(j)}| \leq \varepsilon$ for any $j \in \mathcal{I}_{i_1}(r)$.

Two rooted networks that are not 0-isomorphic are assigned a strictly positive distance inversely related to the largest r and smallest ε such that they have ε -isomorphic r -neighborhoods. Specifically, we define the following distance on the set of rooted networks:

$$d(G_{i_1}, G_{i_2}) := \min \left\{ \inf_{(r, \varepsilon) \in \mathbb{Z}_+ \times \mathbb{R}_{++}} \{ \zeta(r) + \varepsilon : G_{i_1}^r \simeq_\varepsilon G_{i_2}^r \}, 2 \right\}, \quad (1)$$

where $\zeta(x) = (1 + x)^{-1}$. We demonstrate that $d(\cdot, \cdot)$ is a pseudometric in Supplemental Appendix A. The outer minimization is not essential, but taking d to be bounded simplifies later arguments. Further, note that our specific choice of $\zeta(\cdot)$ is arbitrary, in that any function that is supported on $\mathbb{R}_+$, which is monotonically decreasing to zero would be applicable, but we fix our choice of $\zeta(\cdot)$ for concreteness.

The notions of an ε -isomorphism and the rooted network distance d are illustrated in Figure 1. Panels (b) and (c) both depict a pair of rooted networks that are 0-isomorphic and so have a distance of 0. Panel (d) depicts a pair of rooted networks that are not ε -isomorphic for any $\varepsilon > 0$ because the two networks do not have the same number of vertices (and so there cannot exist a bijection between them), but these networks truncated up to a radius of 2 are in fact 0-isomorphic. Other illustrations can be found in Tables 8–10 in Supplemental Appendix D.

We use $\mathcal{G}$ to denote the set of equivalence classes of all possible locally finite rooted networks under d . We demonstrate that $(\mathcal{G}, d)$ is a separable and complete (pseudo) metric space in Supplemental Appendix A.³ Following Aldous and Steele (2004), we call the topology on $\mathcal{G}$ induced by d the local topology, and more broadly call modeling on $\mathcal{G}$ the local approach.

To be sure, the ε -isomorphism used to define the network distance d can be modified to fit the researcher's specific setting. For instance, economic theory may suggest that two agents are qualitatively similar if their rooted networks are similar under edit distance (i.e., they can be made ε -isomorphic by adding or deleting a small number of agents or links). In this case, the researcher may wish to relax the ε -isomorphism to allow for such discrepancies. The results developed in Section 3 continue to hold mutatis mutandis with alternative distance metrics. We use the specific metric (1) because it was the simplest version we could think of such that any two agents with a distance of 0 are observationally equivalent. That is, there is no remaining information in the treatment assignments and network connections that can be used to distinguish them.

2. ECONOMETRIC MODEL

2.1 Main specification

Each agent $i \in \mathcal{I}$ has an outcome $Y_i \in \mathbb{R}$. The population vector of outcomes is denoted $\mathbf{Y} = \{Y_i\}_{i \in \mathcal{I}}$. Our main specification is

$$Y_i = h(G_i(D, \mathbf{T}), U_i). \quad (2)$$

In this specification, Y_i is determined by three factors: the population treatment assignments $\mathbf{T}$ (adding covariates to the model is straightforward, see Appendix A.1), the population network connections D , and agent-specific policy-invariant unobserved heterogeneity term U_i . In some settings, for instance the empirical illustration we consider in Section 4, the outcome model will depend solely on the network configuration D , and thus the treatment assignment vector $\mathbf{T}$ would not appear in model specification (2); while our empirical illustration does not have node level treatments, we do study the performance of our approach in a model with node level treatments using simulations in Appendix Section C.3.

³In principle, our definition of a network and the corresponding definition of $d(\cdot, \cdot)$ could be modified to allow for edges, which are continuously weighted. However, our proof of completeness would no longer be valid in this case.

The function $h : \mathcal{G} \times \mathcal{U} \rightarrow \mathbb{R}$ maps G_i and U_i into the outcome Y_i .⁴ We assume that the researcher has some knowledge of $\mathbf{Y}$, $\mathbf{T}$, and D . U_i and h are unknown. The key assumption of the model is that the effect of the first two factors on the outcome is intermediated by the rooted network $G_i(D, \mathbf{T}) : \mathcal{D} \times \mathcal{T} \rightarrow \mathcal{G}$. That is, for a fixed level of unobserved heterogeneity U_i , any pair of treatment assignments and network connections $(d, \mathbf{t}) \in \mathcal{D} \times \mathcal{T}$ that result in the same rooted network $G_i(d, \mathbf{t})$ lead to the same outcome $h(G_i(d, \mathbf{t}), U_i)$. In the network interference literature, a function of the treatment assignments and network connections that has this property is often called an *exposure map* and any element of the range of the exposure map is called an *effective treatment*.

What distinguishes our specification from the established network interference literature is that, in the literature, the exposure map is typically taken to be low-dimensional, known up to a finite-dimensional vector of parameters. For instance, the exposure map may be a linear function of the number of treated agents with a certain distance to the focal agent. A drawback of this strategy is that the researcher may misspecify the exposure map, that is, they choose a low-dimensional function that does not fully intermediate the effect of the treatment assignments and network connections on the outcome. When the exposure map is misspecified, the researcher's inferences about the impact of policy counterfactuals will not generally be valid. For a further discussion of this point, see our note [Auerbach, Auerbach, and Tabord-Meehan \(2024\)](#).

Our model, by contrast, takes the exposure map to be rooted-network valued. A benefit of this approach is that the rooted network contains everything that is knowable about an agent based on the nearby treatment assignments and network connections. Because of this, it nests a large class of potential exposure maps and, as a result, is less prone to misspecification. A cost of this approach is that, in many settings, the space of rooted networks may be relatively complicated, potentially leading to slow rates of convergence. We discuss this issue in more detail in Section 4 below.

2.2 Examples

We illustrate the prevalence of our main specification (2) with three concrete examples from the network economics literature. In each example, it is possible to specify a valid low-dimensional exposure map that summarizes the information in the treatment assignments and network connections that is relevant for each agent's outcome. However, the natural exposure map in one example is not valid in the setting of another. Our rooted-network-valued exposure map, by contrast, is valid in all three settings.

EXAMPLE 1 (Neighborhood spillovers). Agents are assigned to either treatment or control status with $T_i = 1$ if i is treated and $T_i = 0$ if i is not. Agent i 's outcome depends on their treatment status and the number of treated agents nearby

$$Y_i = Y(T_i, T_i(r), U_i),$$

⁴ U_i is assumed to take values in a separable metric space $\mathcal{U}$. We endow $\mathcal{G} \times \mathcal{U}$ with the usual product topology, define probability measures on the corresponding Borel sigma-algebra, and associate a stochastic rooted network and error pair (G_i, U_i) with a probability measure μ . For now, we take μ as arbitrary and fixed by the researcher. We motivate a specific choice of μ in the context of a multiple-networks setting in Section 3.2 below.

where $T_i(r) = \sum_{j \in \mathbb{N}} T_j \mathbb{1}\{\rho(i, j) \leq r\}$ is the neighborhood count of treatment assigned within some fixed radius r of i ; see, for instance, [Cai, De Janvry, and Sadoulet \(2015\)](#), [Leung \(2016\)](#), [He and Song \(2018\)](#), [Viviano \(2019\)](#). We recast this model as a special case of (2) by noting that both T_i and $T_i(r)$ can be written as functions of G_i , agent i 's rooted network. Specifically, T_i is associated with agent i and so is determined by G_i^0 . $T_i(r)$ counts the number of treated agents within distance r of i and so is determined by G_i^r . It follows that $Y_i = h(G_i, U_i)$ for some h .

EXAMPLE 2 (Social capital formation). Agents leverage their network connections to garner favors, loans, advice, etc. [Jackson, Rodriguez-Barraquer, and Tan \(2012\)](#) specify a model in which one agent performs a favor for another when there is a third agent connected to both that can monitor the exchange. We consider a stochastic version of this model where agent i 's stock of social capital depends on their number of monitored connections

$$Y_i = \left(\sum_{j \in \mathcal{I}} \mathbb{1}\{\mathcal{I}_i(1) \cap \mathcal{I}_j(1) \neq \emptyset\} \right) \cdot U_i.$$

[Karlan, Mobius, Rosenblat, and Szeidl \(2009\)](#), [Cruz, Labonne, and Querubin \(2017\)](#) specify related models of social capital. An example policy of interest is the effect of adding additional connections to the network. We recast this model as a special case of (2) by noting that the number of agents of path distance 2 from i , $\sum_{j \in \mathcal{I}} \mathbb{1}\{\mathcal{I}_i(1) \cap \mathcal{I}_j(1) \neq \emptyset\}$, is determined by G_i^2 . It follows that $Y_i = h(G_i, U_i)$ for some h .

EXAMPLE 3 (Social interactions). Agent i 's equilibrium outcome depends on the number and length of paths between them and the other agents, the treatment statuses of those other agents, and the average treatment statuses of the agents nearby those other agents

$$Y_i = \lim_{S \rightarrow \infty} \sum_{s=0}^S [\delta^s A^*(D)^s (\mathbf{T}\beta + \mathbf{T}^*(1)\gamma)]_i + U_i,$$

where $T_i^*(1) = T_i(1)/N_i(1)$, $[\cdot]_i$ is the i th entry of a vector, and the ij th entry of $A^*(D)$ is $A_{ij}^*(D) = \mathbb{1}\{0 < D_{ij} \leq 1\}/N_i(1)$; see, for instance, [Bramoullé, Djebbari, and Fortin \(2009\)](#), [Blume, Brock, Durlauf, and Ioannides \(2010\)](#), [Giorgi, Pellizzari, and Redaelli \(2010\)](#), [Lee, Liu, and Lin \(2010\)](#), [Goldsmith-Pinkham and Imbens \(2013\)](#). An example policy effect of interest is the average effect of removing a particular agent from the community; see, for instance, [Ballester, Calvó-Armengol, and Zenou \(2006\)](#), [Calvó-Armengol, Patacchini, and Zenou \(2009\)](#), [Lee, Liu, Patacchini, and Zenou \(2021\)](#). We recast this model as a special case of (2) by noting that the i th entry of $\delta^s A^*(D)^s (\mathbf{T}\beta + \mathbf{T}^*(1)\gamma)$ only depends on the treatment statuses and connections of agents within path-distance $s + 1$ of i and so is determined by G_i^{s+1} . It follows that $Y_i = \lim_{S \rightarrow \infty} \sum_{s=1}^S h_s(G_i^s, U_i) = h(G_i, U_i)$ for some functions h_s and h .

There are two key differences between Example 3 and Examples 1 and 2. First, in Example 3, the effective treatment for i depends on all of the treatment statuses and network connections of the agents path connected to i . Leung (2019) argues that this is a common feature of many economic models of network interference. Second, in Example 3, the description of the effective treatment depends on model parameters, which are unknown to the policymaker. Our local approach is, to our knowledge, the first to accommodate these two features when modeling causal effects under network interference.

To be sure, in each of the three examples, it is possible to take the exposure map to be a relatively simple function of the treatment assignments and network connections. For instance, in Example 1, a valid exposure map is $(T_i, T_i(r))$, and in Example 3 it is $(\{[A^*(D)^s(\mathbf{T}\beta + \mathbf{T}^*(1))]\}_{s \in \mathbb{N}}\})$. However, the exact choice of exposure map requires the researcher to commit to one parametric specification. Our approach, by contrast, results in a valid exposure map for all three examples.

2.3 Parameters of interest

Our main objects of interest are the average structural function (ASF) and distributional structural function (DSF) that describe the outcome for i associated with a policy that sets the rooted network to some $g \in \mathcal{G}$. These are, respectively,

$$h(g) = E[h(g, U_i)] \quad \text{and} \quad h_y(g) = E[\mathbb{1}\{h(g, U_i) \leq y\}], \quad (3)$$

where the expectation refers to the marginal distribution of U_i under μ (see footnote 4 in Section 2.1 above); see, for instance, Blundell and Powell (2003). These functions can be used to estimate and conduct inferences about many causal effects of interest. For example, the average treatment effect (ATE) associated with a policy that alters agent i 's rooted network from g to g' is described by $h(g') - h(g)$. Under appropriate continuity assumptions, $h(g)$ and $h_y(g)$ can be approximated by averaging the outcomes corresponding to rooted networks that are close to g under d . We demonstrate this idea with applications to estimating policy effects/treatment response and testing policy irrelevance/no treatment effects in Sections 3.2 and 3.3 below.

This ATE is related to the average exposure effect of Leung (2019), Sävje, Aronow, and Hudgens (2021). These authors consider inference in the setting where the exposure map chosen by the researcher is potentially invalid. A limitation of their approach is that an exposure effect based on a misspecified exposure map may not characterize any policy of interest. We limit the scope for misspecification by using a general class of exposure maps indexed by the space of rooted networks.

We could alternatively consider conditional treatment effect parameters. For instance, the researcher may be interested in a policy that alters treatment assignment conditional on the structure of the network. Or the policy may alter the network connections conditional on some observed agent attributes. In such cases, the researcher may wish to consider a conditional average treatment effect where the researcher conditions on the part of the rooted network not altered by the policy. Such parameters may in some cases be identified under alternative assumptions than those required to identify the above unconditional ATE. This is discussed following Assumption 1 below.

3. APPLICATIONS TO CAUSAL INFERENCE

We apply the local approach framework of Section 2 to two causal inference problems. In both problems, the researcher begins with a rooted network associated with a status-quo policy and is tasked with evaluating the impact of an alternative. The researcher has access to data on outcomes and policies from multiple independent communities or clusters such as schools, villages, firms, or markets. This data structure, described in Section 3.1, is not crucial to our methodology, but simplifies the analysis. It is also very common in the network economics literature. We leave applications to alternative settings, for example, with data from one large network or endogenous policies, to future work.

Our first application, described in Section 3.2, is to the estimation and inference of policy effects. We construct a k -nearest neighbors estimator for the average structural function in (3) and provide nonasymptotic bounds on its estimation error. We then establish an asymptotic normality result for the estimator that can be used to construct confidence intervals for the policy effect in the usual way. Our second application, described in Section 3.3, is to testing policy irrelevance. That is, we test the hypothesis that the rooted networks associated with two policies generate the same distribution of outcomes. These applications contrast a literature studying the magnitude of any potential spillover effects or testing the hypothesis of no spillovers (see, for instance, Aronow (2012), Athey, Eckles, and Imbens (2018), Hu, Li, and Wager (2021), Sävje, Aronow, and Hudgens (2021)).

3.1 Multiple networks setting

A random sample of communities (clusters) are indexed by $c \in [C] := 1, \dots, C$. Each $c \in [C]$ is associated with a finite collection of m_c observations $\{W_{ic}\}_{i \in [m_c]}$ where m_c is a random positive integer, $W_{ic} := (Y_{ic}, G_{ic})$, and $Y_{ic} := h(G_{ic}, U_{ic})$ for some unobserved error U_{ic} . Intuitively, each community c is represented by an initial network connecting m_c agents and the m_c rooted networks refer to this initial network rooted at each agent in the community. Let $W_c := \{W_{ic}\}_{i \in [m_c]}$. We impose the following assumptions on $\{(G_{ic}, U_{ic})\}_{i \in [m_c], c \in [C]}$.

ASSUMPTION 1.

- (i) $\{W_c\}_{c \in [C]}$ are independent and identically distributed (across communities).
- (ii) $\{U_{ic}\}_{i \in [m_c], c \in [C]}$ are identically distributed (within and across communities).
- (iii) For any measurable f , $i \in [m_c]$, and $c \in [C]$,

$$E[f(G_{ic}, U_{ic}) | G_{1c}, \dots, G_{m_c c}] = E[f(G_{ic}, U) | G_{ic}],$$

where U is an independent copy of U_{ic} (i.e., U has the same marginal distribution as U_{ic} but is independent of $\{W_c\}_{c \in [C]}$).

Assumption 1(i) is what makes our analysis “multiple networks.” It states that the networks and errors are independent and identically distributed across communities.

We use this independence across communities to characterize the statistical properties of our test procedure and estimator below. We do not make any restrictions on the dependence structure between observations within a community. This allows for arbitrary dependence within a community; see, for instance, Example 3. Weakening the independence assumption (for instance, considering dependent data from one large community) would require additional assumptions about the intracommunity dependence structure, which we leave to future work.

Assumption 1(ii) fixes the marginal distribution of the errors. It is used to define the policy effect of interest. The relevant structural function is defined as the expected outcome over the homogeneous marginal distribution of U_{ic} for a fixed rooted network. This assumption can be dropped, for example, by defining the expectation to be with respect to the (mixture) distribution of $U_{\iota c}$ generated by drawing ι_c uniformly at random from $[m_c]$.

Assumption 1(iii) states that the rooted networks are exogenous (i.e., the errors are policy irrelevant). We require that the conditional distribution of (G_{ic}, U_{ic}) given $G_{1c}, \dots, G_{m_c c}$ is equal to the conditional distribution of (G_{ic}, U) given G_{ic} , where U is an independent copy of U_{ic} . Exogeneity is a strong assumption, but allows us to approximate the unknown policy functions using sample averages.

The literature cited in Section 2.2 provides two separate motivations for assuming that the network is exogenous. One motivation comes from the random assignment of network connections in an experiment. An example of this is [Azoulay, Graff Zivin, and Wang \(2010\)](#). Another motivation comes from correct model specification, where the researcher accounts for all of the relevant determinants of the outcome on the right-hand side of their model. An example of this is [Calvó-Armengol, Patacchini, and Zenou \(2009\)](#). Our Assumption 1 is consistent with both motivations.

The study of endogenous rooted networks where the policy $(\mathbf{T}, D)$ is potentially related to the errors $\mathbf{U}$ is left to future work. If the policymaker is interested in identifying a conditional average treatment effect as described in the discussion following the definition of $h(g)$ in Section 2.3, then Assumption 1(iii) could be modified to a conditional exogeneity assumption where the structural errors are independent of the effective treatments conditional on the relevant part of the rooted network.

3.2 Estimating policy effects

The policymaker begins with a status-quo policy described by a treatment and network pair $(\mathbf{t}, d)$, and proposes an alternative $(\mathbf{t}', d')$. The researcher is tasked with estimating the expected effect of the policy change for an agent whose effective treatment under the status quo is described by the rooted network $g \in \mathcal{G}$ and whose effective treatment under the alternative is described by the rooted network $g' \in \mathcal{G}$. Following Section 3.2, the potential outcomes under policies g and g' are described by $h(g, U)$ and $h(g', U)$, respectively, for some error U . The object of interest is the policy effect given by

$$h(g') - h(g),$$

where $h(g) = E[h(g, U)]$.

We provide two examples illustrating how this policy effect could be used in practice.

EXAMPLE 4 (Cluster-randomized experiment). An individualized binary treatment is randomly assigned at the cluster level (so that every member of the cluster is assigned the same treatment). In this case, we take g' to be the rooted network of a given individual in a community when their cluster is treated, and g to be the counterfactual rooted network with the same agents and connections but with none of the agents treated. The policy effect is then the difference in expected outcomes for the given individual under the two rooted network structures.

EXAMPLE 5 (Removal of an agent). A central agent is removed from their community. In this case, we take g' to be the rooted network of a remaining individual in the community with the central agent removed, and g to be the counterfactual rooted network that existed before the central agent was removed. The policy effect is then the difference in expected outcomes for the remaining individual under the two rooted network structures.

The proposed estimator for the policy effect is described in Section 3.2.1. In words, it compares the average outcome of the k agents whose rooted networks are most similar to g to the analogous average for g' , using the network distance d .

We impose smoothness conditions on the model parameters and bound the variance of the outcome. We do not believe these assumptions to be restrictive in practice.

ASSUMPTION 2. For $g_0 \in \{g, g'\}$, $\psi_{g_0}(\ell) := P(\min_{i \in [m_c]} d(G_{ic}, g_0) \leq \ell)$ is continuous at every $\ell \geq 0$.

The function $\psi_g(\ell)$ measures the probability that there exists an agent from a randomly drawn community whose rooted network is within ℓ of g . Assumption 2 states that $\psi_g(\ell)$ and $\psi_{g'}(\ell)$ are continuous in ℓ . It justifies a probability integral transform used to characterize the bias of the estimator. The assumption can be guaranteed by adding a randomizing component to the metric (see the discussion following equation (19) in Györfi and Weiss (2020)), which allows for G_{ic} to be discrete.

ASSUMPTION 3. For $g_0 \in \{g, g'\}$, there exists an increasing function $\phi_{g_0} : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ such that $\phi_{g_0}(x) \rightarrow \phi_{g_0}(0) = 0$ as $x \rightarrow 0$ and for every $\tilde{g} \in \mathcal{G}$,

$$|h(g_0) - h(\tilde{g})| \leq \phi_{g_0}(d(g_0, \tilde{g})).$$

Assumption 3 states that h has a modulus of continuity ϕ_{g_0} at g_0 . Such a smoothness condition is standard in the nonparametric estimation literature. It is not generally testable. A model of network interference implies a specific choice of ϕ_g . The three examples of Section 2.2 all satisfy the assumption with $\phi_{g_0}(x) = C^{(x-1)/x}$ for some $C > 1$ potentially depending on parameters of the model but not g_0 . See Supplemental Appendix B for details.

ASSUMPTION 4. For every $\tilde{g} \in \mathcal{G}$, $E[h(\tilde{g}, U_{ic})^2] \leq \bar{\sigma}^2$ for some $\bar{\sigma}^2 > 0$.

Assumption 4 bounds the variance of the outcome variable. It is also standard in the nonparametric estimation literature.

3.2.1 Estimator Let $D_c(g) := \min_{i \in [m_c]} d(G_{ic}, g)$ be the distance of the closest rooted network to g in network c , and let $\mathcal{I}_c(g) := \arg \min_{i \in [m_c]} d(G_{ic}, g)$ be the corresponding set of roots of the rooted networks that achieve this minimum. For each $c \in [C]$, let $\bar{Y}_c(g) := \frac{1}{|\mathcal{I}_c(g)|} \sum_{i \in \mathcal{I}_c(g)} Y_{ic}$ denote the average outcome in $\mathcal{I}_c(g)$. Order $\{\bar{Y}_c(g)\}_{c \in [C]}$ to be increasing in $D_c(g)$ (where ties are broken uniformly at random: see Assumption 2 above and the corresponding discussion). Let the reordered data sequence be given by $(D_1^*(g), \bar{Y}_1^*(g)), (D_2^*(g), \bar{Y}_2^*(g)), \dots, (D_C^*(g), \bar{Y}_C^*(g))$. The proposed estimator for $h(g)$ is then the average of $\{\bar{Y}_k^*(g)\}_{1 \leq j \leq k}$ for some $1 \leq k \leq C$:

$$\hat{h}(g) := \frac{1}{k} \sum_{j=1}^k \bar{Y}_j^*(g)$$

and the analogous estimator for $h(g') - h(g)$ is

$$\hat{h}(g') - \hat{h}(g) := \frac{1}{k} \sum_{j=1}^k (\bar{Y}_j^*(g') - \bar{Y}_j^*(g)).$$

3.2.2 Bound on estimation error We derive a finite-sample bound on the mean-squared error of $\hat{h}(g)$ building on work by Biau and Devroye (2015), Döring, Györfi, and Walk (2017), Györfi and Weiss (2020).

THEOREM 1. Under Assumptions 1, 2, 3, and 4,

$$E[(\hat{h}(g) - h(g))^2] \leq \frac{\bar{\sigma}^2}{k} + E[\varphi_g(U_{(k,C)})^2],$$

where $\varphi_g(x) = \phi_g \circ \psi_g^\dagger(x)$, $\psi_g^\dagger: [0, 1] \rightarrow \mathbb{R}_+$ refers to the upper generalized inverse

$$\psi_g^\dagger(x) = \sup\{\ell \in \mathbb{R}_+ : \psi_g(\ell) \leq x\},$$

and $U_{(k,C)}$ is distributed $\text{Beta}(k, C - k + 1)$.

The bound in Theorem 1 features a familiar bias-variance decomposition. It also establishes the consistency of $\hat{h}(g)$ in well-behaved settings. For example, if (for g fixed) $\varphi_g(\cdot)$ is bounded, continuous at zero, $\varphi_g(0) = 0$, and $k \rightarrow \infty$, $k/C \rightarrow 0$ as $C \rightarrow \infty$ then

$$\bar{\sigma}^2/k \rightarrow 0 \text{ and } E[\varphi_g(U_{(k,C)})^2] = E[(\phi_g \circ \psi_g^\dagger(U_{(k,C)}))^2] \rightarrow 0.$$

The variance component $\bar{\sigma}^2/k$ is standard and decreases as k grows large. In contrast, the bias component $E[\varphi_g(U_{(k,C)})^2]$ and its relationship with k and C are difficult to characterize without further information about ϕ_g and ψ_g . Intuitively, the first controls the

smoothness of the regression function $h(g)$ and the second controls the proximity of the nearest neighbors that make up $\hat{h}(g)$ in terms of proximity to g . We provide further discussion of how the bound depends on features of these parameters in Supplemental Appendix B. We note that the bound remains valid even if we do not observe rooted networks that are arbitrarily close to g as C grows; in this case, we would not expect $\hat{h}(g)$ to be consistent unless $\phi_g(\cdot)$ is exactly zero outside of a neighborhood of g . Theorem 1 has the immediate corollary.

COROLLARY 1. *Suppose the hypothesis of Theorem 1. Then*

$$E[(\hat{h}(g') - \hat{h}(g) - (h(g') - h(g)))^2] \leq \frac{4\bar{\sigma}^2}{k} + 4E[\varphi_{g \vee g'}(U_{(k,C)})^2],$$

where $\varphi_{g \vee g'} := \varphi_g \vee \varphi_{g'}$.

3.2.3 Limiting distribution of $\hat{h}(g) - \hat{h}(g')$ We derive the limiting distribution of $\hat{h}(g)$ (and $\hat{h}(g) - \hat{h}(g')$) under specific assumptions on the rate of growth of the nearest neighbors parameter k relative to C . We impose the following additional assumptions on the data generating process.

ASSUMPTION 5. *For any $\ell > 0$ and $g_0 \in \{g, g'\}$, $\psi_{g_0}(\ell) := P(\min_{i \in [m_C]} d(G_{ic}, g_0) \leq \ell) > 0$.*

Recall that $\psi_g(\ell)$ measures the probability that there exists an agent from a randomly drawn community whose rooted network is within distance ℓ of g under d . Assumption 5 states that for a randomly drawn community there exists a rooted network within distance ℓ of g or g' with positive probability. It implies that as the number of communities C grows, the researcher will eventually observe rooted networks that are arbitrarily close to g or g' .

ASSUMPTION 6. *For g_0 in $\{g, g'\}$, the conditional means*

$$E[\bar{Y}_c(g_0)|D_c(g_0) = d] \text{ and } E[\bar{Y}_c(g_0)^2|D_c(g_0) = d]$$

are continuous in a neighborhood of $d = 0$. The conditional fourth moment of $\bar{Y}_c(g_0)$ is uniformly bounded, that is,

$$E[\bar{Y}_c(g_0)^4|D_c(g_0) = d] \leq M,$$

for some $M < \infty$, for all d .

Assumption 6 imposes some additional sufficient continuity to guarantee that the conditional variance $\text{var}(\bar{Y}_c(g_0)|D_c(g) = d)$ is well behaved as a function of d . The bound on the conditional fourth moments of $\bar{Y}_c(g_0)$ could be reasonably weakened at the cost of introducing more complicated arguments. We present the limiting distribution of $\hat{h}(g)$ as the number of nearest neighbors k grows large at an appropriate rate.

THEOREM 2. *Maintain Assumptions 1–6. Further, suppose that $k \rightarrow \infty$, $k/C \rightarrow 0$ and $kE[\varphi_g(U_{(k,C)})^2] \rightarrow 0$. Then*

$$\sqrt{k}(\hat{h}(g) - h(g)) \xrightarrow{d} N(0, \sigma_g^2(0)),$$

where $\sigma_g^2(d) := \text{var}(\tilde{Y}_c(g) | D_c(g) = d)$.

We also obtain the following immediate corollary for the limiting distribution of $\hat{h}(g) - \hat{h}(g')$ when the estimators $\hat{h}(g)$ and $\hat{h}(g')$ are computed from disjoint subsamples of $\{W_c\}_{c \in [C]}$.

COROLLARY 2. *Consider a partition of the data $\{W_c\}_{c \in [C]}$ into two disjoint sets labelled $\mathcal{W}_1$ and $\mathcal{W}_2$ of size C_1 and C_2 , respectively. Let $\hat{h}(g)$ be the k -nearest neighbors estimator of $h(g)$ computed on $\mathcal{W}_1$ and let $\hat{h}(g')$ be the k -nearest neighbors estimator of $h(g')$ computed on $\mathcal{W}_2$. Maintain Assumptions 1–6. Further, suppose that $k \rightarrow \infty$, $k/\min\{C_1, C_2\} \rightarrow 0$ and $kE[\varphi_{g \vee g'}(U_{k,C})^2] \rightarrow 0$. Then*

$$\sqrt{k}(\hat{h}(g) - \hat{h}(g') - (h(g) - h(g'))) \xrightarrow{d} N(0, \sigma_g^2(0) + \sigma_{g'}^2(0)).$$

To perform inference on $h(g)$ (or $h(g) - h(g')$) using these results, an estimator of the asymptotic variance $\sigma_{g_0}^2(0)$ for $g_0 \in \{g, g'\}$ can be constructed as

$$\hat{\sigma}_{g_0}^2 := \frac{1}{k} \sum_{j=1}^k (\tilde{Y}_j^*(g_0) - \hat{h}(g_0))^2.$$

Theorem 3 establishes conditions under which this estimator is consistent.

THEOREM 3. *Maintain Assumptions 1–6. Further, suppose that $k \rightarrow \infty$ and $k/C \rightarrow 0$, then for $g_0 \in \{g, g'\}$*

$$\hat{\sigma}_{g_0}^2 \xrightarrow{P} \sigma_{g_0}^2(0).$$

Combining Theorem 2 (or Corollary 2) with Theorem 3 establishes the asymptotic validity of inferences based on these estimators.

REMARK 1. Our results are derived under the common assumption that k grows at such a rate that we achieve “undersmoothing.” Intuitively, this means that k must grow sufficiently slowly to guarantee that the bias induced by choosing nearest neighbors “far away” from g is not too large. Although this condition may be straightforward to satisfy in certain special cases (for instance, if the smoothness function $\phi_g(\cdot)$ is suspected to decay very quickly to zero), in general it introduces a tension in the choice of k when using Theorem 2 to conduct inference on $h(g)$; we want k to be large enough to guarantee that the normal approximation is appropriate, while being small enough to ensure that we do not introduce an asymptotic bias into the limiting distribution. We could consider procedures, which explicitly account for this asymptotic bias (see, for instance, [Armstrong](#)

and Kolesár (2020)). However, we leave the study of appropriate modifications of these approaches to our setting to future work. Instead, in Section 3.3 we propose a test for a closely related but distinct hypothesis of policy irrelevance, which states that the rooted networks associated with two policies generate the same distribution of outcomes. As we will show, the gain from testing this stronger null hypothesis is that the resulting test will be asymptotically valid even when the number of nearest neighbors is held fixed in the limit.

REMARK 2. Our results concern the policy effect $h(g') - h(g)$ where g' and g are both specific elements of $\mathcal{G}$. In many settings, however, the policies might refer to a collection of networks $\mathcal{G}_1$ and $\mathcal{G}_2$ where $\mathcal{G}_1, \mathcal{G}_2 \subset \mathcal{G}$, that is, the parameter of interest is $E[h(G_i, U_i)|G_i \in \mathcal{G}_1] - E[h(G_i, U_i)|G_i \in \mathcal{G}_2]$. For example, $\mathcal{G}_1$ may be the set of rooted networks that are observed in particular community, or the set of rooted networks where exactly half of the root's neighbors are treated. It should be possible to extend our methodology to these cases by first estimating the conditional expectation of the outcome with respect to each rooted network in each collection, and then taking a weighted average of the results to estimate $E[h(G_i, U_i)|G_i \in \mathcal{G}_1]$ and $E[h(G_i, U_i)|G_i \in \mathcal{G}_2]$. Following the literature on two-step semiparametric estimation (see, broadly, Powell (1994)), it is likely that, under certain conditions, this double averaging would lead to faster rates of convergence than those described in Theorems 1 and 2 above.

3.3 Testing policy irrelevance

The policy maker begins with a status-quo community policy described by a treatment and network pair (t, d) , and proposes an alternative (t', d') . The researcher is tasked with testing whether the two policies are associated with the same distribution of outcomes for an agent whose effective treatment under the status quo is described by the rooted network $g \in \mathcal{G}$ and whose effective treatment under the alternative is described by the rooted network $g' \in \mathcal{G}$. One can extend our procedure to test policy irrelevance for multiple agents via a multiple testing procedure.

Following Section 3.2, the potential outcomes under g and g' are given by $h(g, U)$ and $h(g', U)$, respectively, for some error U . The hypothesis of policy irrelevance is

$$H_0 : h_y(g) = h_y(g') \quad \text{for every } y \in \mathbb{R}, \quad (4)$$

where $h_y(g) := E[\mathbb{1}\{h(g, U) \leq y\}]$. Under Assumption 1(iii), $h_y(g)$ describes the conditional distribution of Y_i given $G_i \simeq g$.

We revisit the two examples from Section 3.2 to illustrate how this hypothesis test may be used in practice.

EXAMPLE 6 (Cluster-randomized experiment, continued). Recall that in this example an individualized binary treatment is randomly assigned at the cluster-level, g' is the rooted network of a given individual in a community when their cluster is treated, and g is the counterfactual rooted network when their cluster is not treated. The hypothesis of policy irrelevance is that the distribution of outcomes associated with both rooted

networks is the same. A test of this hypothesis may be used to establish that the effect of the treatment is statistically significant: the observed outcomes cannot be explained solely by variation in the unobserved heterogeneity terms.

EXAMPLE 7 (Removal of an agent, continued). Recall that in this example a central agent is removed from their community, g' is the rooted network of a remaining individual in the community with the central agent is removed, and g is the counterfactual rooted network that existed before the central agent was removed. The hypothesis of policy irrelevance is that the distribution of outcomes associated with both rooted networks is the same. A test of this hypothesis may be used to establish that the effect of removing the central agent is statistically significant: the observed outcomes cannot be explained solely by variation in the unobserved heterogeneity terms.

Our test procedure is described in Section 3.3.1. Intuitively, it compares the empirical distribution of outcomes for agents in the data whose rooted networks are most similar to g to the analogous distribution for g' . Asymptotic validity follows after imposing an additional continuity assumption on $h_y(g)$.

ASSUMPTION 7. *For every $\tilde{g} \in \mathcal{G}$, the distribution of $h(\tilde{g}, U)$ is either continuous or discrete with finite support. For every $y \in \mathbb{R}$, $h_y(\tilde{g})$ is continuous in $\tilde{g}$.*

Assumption 7 states that the distribution of outcomes associated with agents whose rooted networks are close to $\tilde{g}$ approximate the distribution of outcomes at $\tilde{g}$. These continuity assumptions are satisfied by the three examples of Section 2.2.

3.3.1 Test procedure We propose an approximate permutation test of H_0 building on Canay, Romano, and Shaikh (2017), Canay and Kamat (2018). The test procedure is described in Algorithm 1. We assume that C is even to simplify notation. When determining the closest agent in Step 1 or reordering the vectors in Step 2, ties are broken uniformly at random.

ALGORITHM 1. Input: data $\{W_c\}_{c \in [C]}$ partitioned into two disjoint sets of size $C/2$ labeled $\mathcal{W}_1$ and $\mathcal{W}_2$, and parameters $q \leq C/2$, $\alpha \in [0, 1]$. Output: a rejection decision.

1. For every $c \in \mathcal{W}_1$, let $i_c(g) := \operatorname{argmin}_{i \in [m_c]} \{d(G_{ic}, g)\}$ be the agent in c whose rooted network G_{ic} is closest to g and $W_c(g) := (Y_c(g), G_c(g)) := (Y_{i_c(g)c}, G_{i_c(g)c})$ be $i_c(g)$'s outcome and rooted network. Similarly, define $W_c(g') := (Y_c(g'), G_c(g'))$ for every $c \in \mathcal{W}_2$.
2. Reorder $\{W_c(g)\}_{c \in \mathcal{W}_1}$ and $\{W_c(g')\}_{c \in \mathcal{W}_2}$ so that the entries are increasing in $d(G_c(g), g)$ and $d(G_c(g'), g')$, respectively. Denote the first q elements of the reordered $\{W_c(g)\}_{c \in \mathcal{W}_1}$

$$W^*(g) := (W_1^*(g), W_2^*(g), \dots, W_q^*(g)),$$

where $W_c^*(g) = (Y_c^*(g), G_c^*(g))$. Similarly, define $W^*(g')$. Collect the $2q$ outcomes of $W^*(g)$ and $W^*(g')$ into the vector

$$S_C := (S_{C,1}, \dots, S_{C,2q}) := (Y_1^*(g), \dots, Y_q^*(g), Y_1^*(g'), \dots, Y_q^*(g')).$$

3. Define the Cramer von Mises test statistic

$$R(S_C) = \frac{1}{2q} \sum_{j=1}^{2q} (\hat{F}_1(S_{C,j}; S_C) - \hat{F}_2(S_{C,j}; S_C))^2,$$

where

$$\hat{F}_1(y; S_C) = \frac{1}{q} \sum_{j=1}^q \mathbb{1}\{S_{C,j} \leq y\} \quad \text{and} \quad \hat{F}_2(y; S_C) = \frac{1}{q} \sum_{j=q+1}^{2q} \mathbb{1}\{S_{C,j} \leq y\}.$$

4. Let $\mathbf{H}$ be the set of all permutations $\pi = (\pi(1), \dots, \pi(2q))$ of $\{1, \dots, 2q\}$ and

$$S_C^\pi = (S_{C,\pi(1)}, \dots, S_{C,\pi(2q)}).$$

Reject H_0 if the p -value $p \leq \alpha$ where $p := \frac{1}{|\mathbf{H}|} \sum_{\pi \in \mathbf{H}} \mathbb{1}\{R(S_C^\pi) \geq R(S_C)\}$.

The p -value in Step 4 may be difficult to compute when $\mathbf{H}$ is large. Theorem 4 below continues to hold if $\mathbf{H}$ is replaced by $\hat{\mathbf{H}}$ where $\hat{\mathbf{H}} = \{\pi_1, \dots, \pi_B\}$, π_1 is the identity permutation and $\pi_2, \dots, \pi_B$ are drawn independently and uniformly at random from $\mathbf{H}$. Such sampling is standard in the literature; see also [Canay and Kamat \(2018\)](#), Remark 3.2.

In practice, we partition $\{W_c\}_{c \in [C]}$ into two disjoint sets iteratively in the following way. We first select the community, which contains the observation closest to g , add it to $\mathcal{W}_1$, and remove it from the pool of candidate communities. We then select the community, which contains the observation closest to g' , add it to $\mathcal{W}_2$, and remove it from the pool of candidate communities. We continue this process, alternating between g and g' , until the pool of candidate communities is exhausted.

Since ties are broken uniformly at random when determining the closest agent in Step 1 or reordering the vectors in Step 2, it is likely that our procedure introduces uncertainty in the p -value. For this reason, we recommend that practitioners assess the sensitivity of these choices on the resulting p -value. Practitioners could also consider formally aggregating several p -values obtained from different choices in Steps 1 and 2 using the methods outlined in Section 2.1 of [DiCiccio, DiCiccio, and Romano \(2020\)](#).

The test presented in Algorithm 1 is nonrandomized in the sense that, once the collection S_C is selected, the decision to reject the null hypothesis is a deterministic function of the data. This leads to a test which is potentially conservative. One could alternatively consider a nonconservative version of this test which is randomized; see [Lehmann and Romano \(2006\)](#), Section 15.2.

3.3.2 Asymptotic validity If the entries of $Y^*(g) = (Y_1^*(g), \dots, Y_q^*(g))$ and $Y^*(g') = (Y_1^*(g'), \dots, Y_q^*(g'))$ were identically distributed to $h(g, U)$ and $h(g', U)$, respectively, then the test described in Algorithm 1 would control size in finite samples following standard arguments (e.g., [Lehmann and Romano \(2006\)](#), Theorem 15.2.1). However, since the rooted networks corresponding to $Y_j^*(g)$ and $Y_j^*(g')$ are not exactly g and g' , such an argument can not be directly applied.

Our assumptions instead imply that in an asymptotic regime where q is fixed and $C \rightarrow \infty$, the entries of $Y^*(g)$ and $Y^*(g')$ are approximately equal in distribution to

$h(g, U)$ and $h(g', U)$. A fixed q rule is appropriate in our setting because the quality of the nearest neighbors to g or g' may degrade rapidly with q . Our simulation evidence in Supplemental Appendix C.2 suggests that setting $q = 2\lfloor \log(C) \rfloor$ works well in practice.

We demonstrate that the test procedure in Algorithm 1 is asymptotically valid building on work by Canay, Romano, and Shaikh (2017), Canay and Kamat (2018). Finite-sample behavior is examined via simulation in Supplemental Appendix C.

THEOREM 4. *Under Assumptions 1, 5, and 7, the test described in Algorithm 1 is asymptotically ($C \rightarrow \infty$, q fixed) level α .*

4. EMPIRICAL ILLUSTRATION

We illustrate our local approach with a study of the influence of network structure on favor exchange in the setting of Jackson, Rodriguez-Barraquer, and Tan (2012). In their work, Jackson, Rodriguez-Barraquer, and Tan (2012) specify an equilibrium model of favor exchange where one agent is willing to perform a favor for another if there is a third agent who has a social connection to both agents and can monitor the exchange, that is, the first two agents are indirectly connected. Intuitively, the social norm is for agents to be altruistic and provide favors for each other and the third agent exerts social pressure on the other two agents to conform to this norm. Without this social pressure, agents have an incentive to ignore the social norm and not provide favors for each other. When such a norm enforcing third agent exists they say that the favor exchange relationship between the first two agents is supported. That is, formally, the relationship between i and j is supported if and only if there exists a $k \in \mathcal{I}$ such that $D_{ik} = D_{jk} = 1$.⁵ The notion of support is extended to agents by counting the number of supported links adjacent to that agent and we refer to this statistic as agent support. That is, formally, agent i 's support is given by $\sum_{j \neq i} \mathbb{1}\{\sum_k D_{ik} D_{jk} > 0\}$.

Support contrasts alternative measures of social cohesion such as clustering. An agent's clustering coefficient measures the fraction of agent pairs connected to the agent that are also connected to each other, that is, $(\sum_{i \neq j \neq k} D_{ij} D_{ik} D_{jk}) / (\sum_{i \neq j \neq k} D_{ij} D_{ik} \vee 1)$. One might think that high levels of agent clustering also drives favor exchange between agents. However, in the Jackson, Rodriguez-Barraquer, and Tan (2012) model, the two are unrelated conditional on support. Intuitively, the decision for agents to exchange favors is made on the extensive margin in that it only depends on whether or not their relationship is monitored. Adding additional monitors may increase the level of clustering, but under the theory does not impact favor exchange all else equal. The authors take the existence of low-clustering, high-support favor exchange networks in real-world network data as corroborating evidence for their model.

⁵This definition of support is given by Jackson, Rodriguez-Barraquer, and Tan (2012) on page 1877 before the statement of Theorem 3. Strictly speaking, the authors only define support for pairs of agents that exchange favors. That is, they say that the relationship between two agents is supported if they exchange favors and their relationship is monitored. We instead say that a relationship is supported if it is monitored, regardless of whether or not the relevant agents exchange favors.

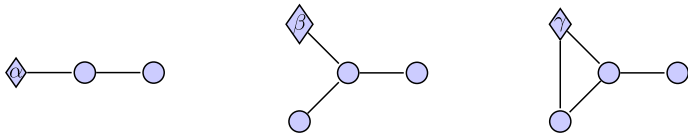


FIGURE 2. These three networks constitute the “cutlery ensemble.” We use them to describe three different ways that agents can monitor each other in a social network. The first network on the left is the “knife” network rooted at agent α . The second network in the middle is the “fork” network rooted at agent β . The third network on the right is the “spoon” network rooted at the agent γ .

We apply our local approach to evaluate the influence of support and clustering on the number of favors exchanged directly. There is no treatment assigned in this illustration, so we take $\mathcal{T}$ to be the empty set. Instead, we compare the average outcomes for collections of agents whose rooted networks are isomorphic to certain fixed configurations with various amounts of support and clustering described in Figure 2. The advantage of our approach relative to the empirical analysis of [Jackson, Rodriguez-Barraquer, and Tan \(2012\)](#) is that we do not assume that the social network only impacts the amount of favors exchanged by an agent through the level of support and clustering. Instead, the outcomes can depend on any feature of the social connections, as determined by the agent’s rooted network. Following [Jackson, Rodriguez-Barraquer, and Tan \(2012\)](#), we consider data from 75 rural villages in Karnakata, India, and construct our favor exchange variable (the outcome) and monitoring networks as described in their footnote 39.⁶ Summary statistics on the number of households in each village are provided in Table 1.

To be sure, the social network links are not randomly assigned in this setting and our causal interpretation hinges on the validity of our Assumption 1. In particular, we assume that any other driver of favor exchange in the village are unrelated to the structure of the monitoring network. To the extent that this exogeneity assumption is violated, the associations inferred from our methodology may not be causal.

Below we estimate a policy effect and test the hypothesis of policy irrelevance for two pairs of rooted networks describing three different ways that agents can monitor each other in the hedonic network. The networks we chose are given by the “cutlery ensemble” of rooted networks depicted in Figure 2. In the knife network, the root agent

TABLE 1. Summary statistics for the number of households per village.

Total Num.	Mean	Min	25% Quantile	50% Quantile	75% Quantile	Max
13,405	179	70	140	167	218	326

⁶Specifically, to construct the monitoring network we use their hedonic network, which documents friends and visitors. To define the number of favors exchanged, we use the number of households in the village to which they report borrowing or lending money or kerorice. The data was originally collected by [Banerjee, Chandrasekhar, Duflo, and Jackson \(2013\)](#) to study the diffusion of information about a microfinance program.

TABLE 2. Estimates and confidence intervals of treatment effects.

		$E[Y_\beta] - E[Y_\alpha]$	$E[Y_\gamma] - E[Y_\beta]$
	Est.	0.24	0.62
$k = 10$	95% CI	$[-0.35, 0.82]$	$[0.21, 1.03]$
	90% CI	$[-0.25, 0.73]$	$[0.27, 0.96]$
	Est.	0.11	0.69
$k = 20$	95% CI	$[-0.20, 0.43]$	$[0.44, 0.95]$
	90% CI	$[-0.15, 0.38]$	$[0.48, 0.91]$
	Est.	0.07	0.74
$k = 30$	95% CI	$[-0.15, 0.29]$	$[0.55, 0.94]$
	90% CI	$[-0.12, 0.26]$	$[0.58, 0.91]$

α has one supported relationship and a clustering coefficient of 0. In the fork network, the root agent β has two supported relationships and a clustering coefficient of 0. In the spoon network, the root agent γ has three supported relationships and a clustering coefficient of 1. We chose these networks for our illustration because they are simple to describe and have varying degrees of support and clustering.

Our first policy comparison is the level of favor exchange in the fork network to that of the knife network. Specifically, we compare the distributions of Y_α and Y_β where the two random variables refer to the number of favors performed by α and β , respectively. Intuitively, this comparison measures the effect of taking α 's community as given by the knife network, adding an additional fourth agent, and connecting them to the center agent of the knife. This policy change increases the support of the root agent from 1 to 2 but does not impact the clustering coefficient of the root. Our second policy comparison is the level of favor exchange in the spoon network to that of the fork network (i.e., we compare the distributions of Y_β and Y_γ). Intuitively, this comparison measures the effect of taking β 's community as given by the fork network and adding a social connection between agent β and the other “prong” of the fork directly below β . This policy change increases the support of the root agent from 2 to 3 and the clustering coefficient of the root agent from 0 to 1. Table 2 reports the point estimates and confidence intervals associated with the average differences in favors for these policy counterfactuals using our k -nearest neighbors estimator. We find that overall moving from fork to spoon leads to a positive and statistically significant increase in the number of favors, whereas this is not the case when moving from knife to fork.

We also test the hypotheses that Y_α and Y_β are equal in distribution, and Y_β and Y_γ are equal in distribution, using the approximate randomization test described in Section 3.3. For the test of $Y_\alpha =_d Y_\beta$, we obtain p-values of 1.000, 1.000, 0.152 for $q = 5, 10, 20$, respectively, which we interpret as providing no evidence against the hypothesis. For the test of $Y_\beta =_d Y_\gamma$, we obtain p-values of 0.062, 0.007, 0.074 for $q = 5, 10, 20$, respectively, so that we have statistical significance at the 10% level.⁷

⁷We note that when performing these tests there exist multiple rooted networks, which are equidistant to the target rooted network both within and across villages; Figures 7-10 in Supplemental Appendix D pro-

Ultimately, we view these empirical results as challenging the view that clustering does not play a role in favor exchange. This is because the first policy comparison (between knife and fork) constitutes an increase in support holding clustering fixed, but we found no evidence of a change in favor exchange. The second policy comparison (between spoon and fork) constitutes both an increase in support and clustering and we did find evidence of an increase in favor exchange. Our conclusion is that there may still be some role for clustering to explain altruism in real-world networks.⁸

5. CONCLUSION

This paper proposes a new nonparametric modeling framework for causal inference under network interference. Rooted networks serve the role of effective treatments in that they index the ways in which the treatment assignments and network structure can influence the agent outcomes. We demonstrate our approach with an estimation strategy for average or distributional policy effects, a test for the hypothesis of policy irrelevance, and an empirical illustration studying the effect of network structure on social capital formation.

Much work remains to be done, as indicated in the discussions of the assumptions and results in the sections above. Other potential directions for future work include considering the problem of policy learning under network interference (see, e.g., [Ananth \(2020\)](#), [Viviano \(2019\)](#), [Kitagawa and Wang \(2020\)](#)), applying the framework to identify and estimate the parameters of a strategic network formation model (as in, for instance, [de Paula, Richards-Shubik, and Tamer \(2018\)](#)), or semiparametrically estimating average treatment effects for binary treatment (as in, for instance, [Abadie and Imbens \(2006\)](#)). Ultimately, we see our work as one step in a direction allowing for the more flexible econometric modeling of sparse network structures.

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vide depictions of these networks. As a consequence, the (arbitrary) selection of such rooted networks introduces additional randomness into the computation of the p -value. Supplemental Appendix D discusses the robustness of our results to these random selections. We additionally note that, beyond the case $q = 5$, the rooted networks matched to knife could equivalently be matched to fork; as a consequence, we may be concerned that the results beyond $q = 5$ are attenuated in this comparison. In contrast, the matches for $q = 5$ are not affected by this issue.

⁸We thank Ben Golub for helping us interpret these results in the context of [Jackson, Rodriguez-Barraquer, and Tan’s \(2012\)](#) model.

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All authors assume responsibility for all aspects of the paper.